Edward F. Poolos
Director

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Deputy Director

Kay Ivey
Governor

Alabama Department of Environmental Management
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(334) 271-7700 ▪ FAX (334) 271-7950

March 17, 2026

Mr. Fred Francis
Senior Environmental Engineer
International Paper - Riverdale Mill
601 County Rd 78
Selma, AL 36703

RE: No. 2 Wood Waste Boilers Compliance Tests and RATA
Facility No. 104-0003

Dear Mr. Francis:

The Department has received and reviewed the particulate matter, carbon monoxide, and sulfur dioxide emissions test report and the NOx and O_2 RATA test for the No. 2 Wood Waste Boiler at your facility in Selma, AL. The testing was performed on February 11, 2026. Enclosed are the results of the Department's evaluation. The Mill has established or confirmed the following parameters:

Source	Parameter	Operational Limit	Units
No. 2 Bark Boiler	Steaming Rate	≤ 413	kpph
No. 2 Bark Boiler	Oxygen	≥ 5.4	%

According to the results, it appears that the units and monitoring systems were operating within the operational limits at the time of the tests.

If you have any questions or comments, please contact me at (334) 270-5671.

Sincerely,

Steven Bissey
Industrial Chemicals Section
Chemical Branch
Air Division

SPB/spb

Enclosure

FILE ID: 6298 104-0003 047 03-17-2026 STR SPB NO 2 BB PM, CO, SO2 & RATA ADR

Birmingham Office
110 Vulcan Road
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1615 South Broad Street
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(251) 479-2593 (FAX)

FACILITY: IP Riverdale
PERMIT NUMBER: 104-0003-X018
EVALUATION BY: HWT
SOURCE: No. 2 Wood Waste Boiler

TEST DATE: 2/11/26
TESTED BY: Weston Solutions
EVALUATION DATE: 3/12/26

EMISSIONS SUMMARY

Run No.	1	2	3	AVERAGE	
Bws	18.1	19.3	21.0	19.4	%
F-factor	9,415	9,368	9,419		dscf/MMbtu
Qstd	140,514	152,179	147,362	146,685	DSCFM
O_2	7.2	6.5	5.1	6.3	%
Particulate Emissions					
PM	0.005	0.011	0.008	0.008	gr/dscf
PM	5.93	13.06	9.25	9.42	lb/hr
PM	**0.01**	**0.02**	**0.01**	**0.02**	**lb/MMbtu**
Allowable				*0.10*	*lb/MMbtu*
CO Emissions					
CO	124.0	59.7	201.3	128.3	ppm
CO	**0.1**	**0.1**	**0.2**	**0.1**	**lb/MMbtu**
Allowable				*0.5*	*lb/MMbtu*
CO	**162**	**74**	**228**	**155**	**ppm @ 3% O_2**
Allowable				*3,500*	*ppm @ 3% O_2*
SO2 Emissions					
SO2	1.1	0.8	0.8	0.9	ppm
SO2	1.5	1.1	1.0	1.22	lb/hr
SO2	**6.6**	**5.0**	**4.43**	**5.34**	**tpy**
Allowable				*38.60*	*tpy*
SO2	**0.0026**	**0.0018**	**0.0016**	**0.0020**	**lb/MMbtu**
Allowable				*0.8*	*lb/MMbtu*

FACILITY NAME: IP Riverdale
PERMIT NUMBER: 104-0003-X018
EVALUATION BY: HWT

TEST DATE: 02/11/26
TESTED BY: Weston Solutions
EVAL. DATE: 3/12/26

SOURCE: #2 Wood Waste Boiler
METHOD 5 PARTICULATE EMISSIONS

RUN #	1	2	3	
Time	0928-1031	1214-1315	1423-1525	
F-Factor	9,415	9,368	9,419	scf/mmbtu
ΔH	1.30	1.32	1.28	"H_2O
ΔP	0.6871	0.6930	0.6861	"H_2O
tm	79	86	85	°F
ts	433	433	439	°F
P_{bar}	29.86	29.86	29.86	"Hg
Pg	-0.84	-0.84	-0.84	"H_2O
Vm	38.503	37.407	35.869	dcf
Y	1.003	1.003	1.003	
CO_2	11.5	12.7	13.8	%
O_2	8.0	6.4	5.4	%
CO	N.A.	N.A.	N.A.	%
Vlc	166.7	174.3	187.7	g
Sg	10.0	9.2	9.1	g
Mn	12.1	25.7	18.1	mg
Cp	0.84	0.84	0.84	
Stack Dia.	132.0	132.0	132.0	inches
dn	0.282	0.282	0.282	inches
time	60	60	60	min

RUN #	1	2	3	AVG	
As	95.033	95.033	95.033		sq. ft.
An	4.34E-04	4.34E-04	4.34E-04		sq. ft.
Ps	29.80	29.80	29.80		"Hg
Pm	29.96	29.96	29.95		"Hg
Md	30.16	30.29	30.42		lb/lb-mole
Vwstd	8.33	8.65	9.28		scf
Vmstd	37.822	36.281	34.888		dscf
Bws	18.1	19.3	21.0	19.4	%
Bws THEO	100	100	100		%
Bws meas.	18.1	19.3	21.0		%
Ms	27.96	27.92	27.81		lb/lb-mole
Vs	51.1	51.6	51.3	51.3	ft/sec
Qa	291,192	293,982	292,642	292,605	acfm
Qstd	140,514	139,724	135,096	138,445	sdcfm
vn	78.4	76.3	75.6	76.7	acf
% ISOKINETIC	98.3	94.8	94.3		
Concentration (Cs)	0.0049	0.0109	0.0080	0.0079	gr/dscf
PM Emissions	5.93	13.06	9.25	9.42	lb/hr
PM Emissions	0.011	0.021	0.014	0.015	lb/MMbtu

FACILITY: IP Riverdale
PERMIT NUMBER: 104-0003-X018
SOURCE: No. 2 Wood Waste Boiler

TEST DATE: 2/11/26
TESTED BY: Weston Solutions
EVALUATION BY: HWT
EVALUATION DATE: 3/12/2026

METHOD 10 MONITOR PERFORMANCE

CO

MONITOR SPAN = 909.00

ACCEPTABLE RANGES				C.G. (%)	TM (%)	% OF SPAN	Acceptable CE ± 2%
LESS THAN		181.8	ZERO	0.00	0.00	0.0	P
363.6	-	545.4	LOW	446.00	446.00	0.0	P
909	-	909	HIGH	909.00	909.00	0.0	P

RUN	INITIAL SYSTEM RESPONSE (%)	UPSCALE INITIAL BIAS (% OF SPAN)	Acceptable BIAS ± 5%	FINAL SYSTEM RESPONSE (%)	FINAL BIAS (% OF SPAN)	Acceptable BIAS ± 5%	MONITOR DRIFT (% OF SPAN)	Acceptable Drift ≤ 3%	INITIAL SYSTEM RESPONSE (%)	ZERO INITIAL BIAS (% OF SPAN)	Acceptable BIAS ± 5%	FINAL SYSTEM RESPONSE (%)	FINAL BIAS (% OF SPAN)	Acceptable BIAS ± 5%	MONITOR DRIFT (% OF SPAN)	Acceptable Drift ≤ 3%
1	446.00	0.0	P	446.00	0.0	P	0.0	P	0.00	0.0	P	0.00	0.0	P	0.0	P
2	446.00	0.0	P	446.00	0.0	P	0.0	P	0.00	0.0	P	0.00	0.0	P	0.0	P
3	446.00	0.0	P	446.00	0.0	P	0.0	P	0.00	0.0	P	0.00	0.0	P	0.0	P
4	446.00	0.0	P	446.00	0.0	P	0.0	P	0.00	0.0	P	0.00	0.0	P	0.0	P
5	446.00	0.0	P	446.00	0.0	P	0.0	P	0.00	0.0	P	0.00	0.0	P	0.0	P
6	446.00	0.0	P	446.00	0.0	P	0.0	P	0.00	0.0	P	0.00	0.0	P	0.0	P
7	446.00	0.0	P	446.00	0.0	P	0.0	P	0.00	0.0	P	0.00	0.0	P	0.0	P
8	446.00	0.0	P	446.00	0.0	P	0.0	P	0.00	0.0	P	0.00	0.0	P	0.0	P
9	446.00	0.0	P	446.00	0.0	P	0.0	P	0.00	0.0	P	0.00	0.0	P	0.0	P

SUMMARY OF RUNS

RUN	Raw CO %-DRY	DRIFT CORR. CO %-DRY	Run Avgs %-DRY		F-factor	O2 %-DRY	CO ppmvd @ 3% O2		lb/MMBtu
1	123.00	123.00							
2	136.00	136.00	124.00	R1	9415	7.21	162.08	R1	0.1295
3	113.00	113.00							
4	45.00	45.00							
5	57.00	57.00	59.67	R2	9368	6.52	74.25	R2	0.0590
6	77.00	77.00							
7	120.00	120.00							
8	137.00	137.00	201.33	R3	9419	5.13	228.48	R3	0.1827
9	347.00	347.00							

*CO ppm @ 3% O2 differs slightly due to calculations from truncated numbers

FACILITY: IP Riverdale
PERMIT NUMBER: 104-0003-X018
SOURCE: No. 2 Wood Waste Boiler

TEST DATE: 2/11/26
TESTED BY: Weston Solutions
EVALUATION BY: HWT
EVALUATION DATE: 3/12/2026

METHOD 6C MONITOR PERFORMANCE

SO2

MONITOR SPAN = 90.30

ACCEPTABLE RANGES				C.G. (%)	TM (%)	% OF SPAN	Acceptable CE ± 2%
LESS THAN		18.06	ZERO	0.00	-0.10	0.1	P
36.12	-	54.18	LOW	45.20	45.50	0.3	P
90.3	-	90.3	HIGH	90.30	90.20	0.1	P

RUN	INITIAL SYSTEM RESPONSE (%)	UPSCALE INITIAL BIAS (% OF SPAN)	Acceptable BIAS ± 5%	FINAL SYSTEM RESPONSE (%)	FINAL BIAS (% OF SPAN)	Acceptable BIAS ± 5%	MONITOR DRIFT (% OF SPAN)	Acceptable Drift ≤ 3%	INITIAL SYSTEM RESPONSE (%)	ZERO INITIAL BIAS (% OF SPAN)	Acceptable BIAS ± 5%	FINAL SYSTEM RESPONSE (%)	FINAL BIAS (% OF SPAN)	Acceptable BIAS ± 5%	MONITOR DRIFT (% OF SPAN)	Acceptable Drift ≤ 3%
1	45.90	0.4	P	44.60	-1.0	P	1.4	P	-0.10	0.0	P	0.20	0.3	P	0.3	P
2	45.90	0.4	P	44.60	-1.0	P	1.4	P	-0.10	0.0	P	0.20	0.3	P	0.3	P
3	45.90	0.4	P	44.60	-1.0	P	1.4	P	-0.10	0.0	P	0.20	0.3	P	0.3	P
4	44.60	-1.0	P	44.60	-1.0	P	0.0	P	0.20	0.3	P	0.00	0.1	P	0.2	P
5	44.60	-1.0	P	44.60	-1.0	P	0.0	P	0.20	0.3	P	0.00	0.1	P	0.2	P
6	44.60	-1.0	P	44.60	-1.0	P	0.0	P	0.20	0.3	P	0.00	0.1	P	0.2	P
7	44.60	-1.0	P	43.80	-1.9	P	0.9	P	0.00	0.1	P	0.20	0.3	P	0.2	P
8	44.60	-1.0	P	43.80	-1.9	P	0.9	P	0.00	0.1	P	0.20	0.3	P	0.2	P
9	44.60	-1.0	P	43.80	-1.9	P	0.9	P	0.00	0.1	P	0.20	0.3	P	0.2	P

SUMMARY OF RUNS

RUN	Raw SO2 %-DRY	DRIFT CORR. SO2 %-DRY	Run Avgs %-DRY		F-factor	FLOW (DSCFM)	O2 %-DRY	lb/hr		TPY	lb/MMBtu
1	1.20	1.15									
2	1.00	0.95	1.08	R1	9415	140514	7.21	1.51615	R1	6.64	0.0026
3	1.20	1.15									
4	1.00	0.91									
5	0.90	0.81	0.81	R2	9368	139724	6.52	1.13084	R2	4.95	0.0018
6	0.80	0.71									
7	0.70	0.61									
8	0.70	0.61	0.75	R3	9419	135096	5.13	1.01135	R3	4.43	0.0016
9	1.10	1.02									

2% of span was used as a detection limit *
2% span = 1.9 ppm

FACILITY: IP Riverdale
PERMIT NUMBER: 104-0003-X018
SOURCE: No. 2 Wood Waste Boiler

TEST DATE: 2/11/2026
TESTED BY: Weston Solutions
EVALUATION BY: HWT
EVALUATION DATE: 3/12/2026

EMISSIONS SUMMARY

RATA GAS	ALLOWABLE RA	BASED ON	RELATIVE ACCURACY	UNITS
O2	1%	Absolute Difference	0.3%	%-DRY

RATA GAS	ALLOWABLE RA	BASED ON	RELATIVE ACCURACY	UNITS
NOx	≤20%	RM	6.8%	lbs/MMBtu

RELATIVE ACCURACY TEST

FACILITY: IP Riverdale
PERMIT NUMBER: 104-0003-X018
SOURCE: No. 2 Wood Waste Boiler

TEST DATE: 2/11/2026
TESTED BY: Weston Solutions
EVALUATION BY: HWT
EVALUATION DATE: 3/12/2026

USE Y/N	RUN	STACK MONITOR %-DRY	O2 REFERENCE %-DRY	DIFF. %-DRY
Y	1	7.00	6.77	-0.23
Y	2	7.90	7.58	-0.32
Y	3	7.60	7.27	-0.33
Y	4	7.60	7.12	-0.48
Y	5	6.20	5.81	-0.39
Y	6	6.80	6.62	-0.18
Y	7	5.70	5.43	-0.27
Y	8	5.80	5.43	-0.37
Y	9	4.80	4.52	-0.28
AVG.		6.60	6.28	-0.32

(n) = NO. OF RUNS	9
STD. DEVIATION	0.09
CONFIDENCE COEFF.	0.07

REL. ACC. BASED ON AVG. DIFF.
RA ALLOWABLE 1 PERCENT **0.32**

FACILITY: IP Riverdale
PERMIT NUMBER: 104-0003-X018
SOURCE: No. 2 Wood Waste Boiler

TEST DATE: 2/11/26
TESTED BY: Weston Solutions
EVALUATION BY: HWT
EVALUATION DATE: 3/12/2026

METHOD 3A MONITOR PERFORMANCE

O2

MONITOR SPAN = **20.01**

ACCEPTABLE RANGES				C.G. (%)	TM (%)	% OF SPAN	*Acceptable CE ± 2%*
LESS THAN		4.002	ZERO	0.00	0.00	0.0	*P*
8.004	-	12.006	LOW	10.00	10.00	0.0	*P*
20.01	-	20.01	HIGH	20.01	20.01	0.0	*P*

RUN	INITIAL SYSTEM RESPONSE (%)	UPSCALE INITIAL BIAS (% OF SPAN)	*Acceptable BIAS ± 5%*	FINAL SYSTEM RESPONSE (%)	FINAL BIAS (% OF SPAN)	*Acceptable BIAS ± 5%*	MONITOR DRIFT (% OF SPAN)	*Acceptable Drift ≤ 3%*	ZERO INITIAL SYSTEM RESPONSE (%)	ZERO INITIAL BIAS (% OF SPAN)	*Acceptable BIAS ± 5%*	FINAL SYSTEM RESPONSE (%)	FINAL BIAS (% OF SPAN)	*Acceptable BIAS ± 5%*	MONITOR DRIFT (% OF SPAN)	*Acceptable Drift ≤ 3%*
1	10.00	0.0	*P*	10.00	0.0	*P*	0.0	*P*	0.10	0.5	*P*	0.10	0.5	*P*	0.0	*P*
2	10.00	0.0	*P*	10.00	0.0	*P*	0.0	*P*	0.10	0.5	*P*	0.10	0.5	*P*	0.0	*P*
3	10.00	0.0	*P*	10.00	0.0	*P*	0.0	*P*	0.10	0.5	*P*	0.10	0.5	*P*	0.0	*P*
4	10.00	0.0	*P*	9.90	-0.5	*P*	0.5	*P*	0.10	0.5	*P*	0.00	0.0	*P*	0.5	*P*
5	10.00	0.0	*P*	9.90	-0.5	*P*	0.5	*P*	0.10	0.5	*P*	0.00	0.0	*P*	0.5	*P*
6	10.00	0.0	*P*	9.90	-0.5	*P*	0.5	*P*	0.10	0.5	*P*	0.00	0.0	*P*	0.5	*P*
7	9.90	-0.5	*P*	9.90	-0.5	*P*	0.0	*P*	0.00	0.0	*P*	0.10	0.5	*P*	0.5	*P*
8	9.90	-0.5	*P*	9.90	-0.5	*P*	0.0	*P*	0.00	0.0	*P*	0.10	0.5	*P*	0.5	*P*
9	9.90	-0.5	*P*	9.90	-0.5	*P*	0.0	*P*	0.00	0.0	*P*	0.10	0.5	*P*	0.5	*P*

SUMMARY OF RUNS

RUN	Raw O2 %-DRY	BIAS CORR. O2 %-DRY	Run Avgs %-DRY	
1	6.80	6.8		
2	7.60	7.6	7.21	R1
3	7.30	7.3		
4	7.10	7.1		
5	5.80	5.8	6.52	R2
6	6.60	6.6		
7	5.40	5.4		
8	5.40	5.4	5.13	R3
9	4.50	4.5		

RELATIVE ACCURACY TEST

FACILITY: IP Riverdale
PERMIT NUMBER: 104-0003-X018
SOURCE: No. 2 Wood Waste Boiler

TEST DATE: 2/11/26
TESTED BY: Weston Solutions
EVALUATION BY: HWT
EVALUATION DATE: 3/12/2026

USE Y/N	RUN	STACK MONITOR lb/MMBtu	NOx MONITOR lb/MMBtu	DIFF.
Y	1	0.260	0.242	-0.018
Y	2	0.280	0.259	-0.021
Y	3	0.273	0.253	-0.020
Y	4	0.261	0.251	-0.010
Y	5	0.236	0.239	0.003
Y	6	0.234	0.235	0.001
Y	7	0.219	0.212	-0.007
Y	8	0.213	0.209	-0.004
Y	9	0.195	0.198	0.003
AVG		0.241	0.233	-0.008
(n) = NO. OF RUNS				9
STD. DEVIATION				0.0098
CONFIDENCE COEFF.				0.0076
REL. ACC. BASED ON RM				
RA% ALLOWABLE		**20**		**6.8**

FACILITY: IP Riverdale
PERMIT NUMBER: 104-0003-X018
SOURCE: No. 2 Wood Waste Boiler

TEST DATE: 2/11/26
TESTED BY: Weston Solutions
EVALUATION BY: HWT
EVALUATION DATE: 3/12/2026

METHOD 7E MONITOR PERFORMANCE

NOx

MONITOR SPAN = **251.00**

ACCEPTABLE RANGES				C.G. (%)	TM (%)	% OF SPAN	*Acceptable CE ± 2%*
LESS THAN		50.2	ZERO	0.00	-1.00	0.4	*P*
100.4	-	150.6	LOW	123.00	126.00	1.2	*P*
251	-	251	HIGH	251.00	250.00	0.4	*P*
Nox Converter Criteria >90.0%				49.60	49.30	99.40	P

RUN	INITIAL SYSTEM RESPONSE (%)	UPSCALE INITIAL BIAS (% OF SPAN)	*Acceptable BIAS ± 5%*	FINAL SYSTEM RESPONSE (%)	FINAL BIAS (% OF SPAN)	*Acceptable BIAS ± 5%*	MONITOR DRIFT (% OF SPAN)	*Acceptable Drift ≤ 3%*	INITIAL SYSTEM RESPONSE (%)	ZERO INITIAL BIAS (% OF SPAN)	*Acceptable BIAS ± 5%*	FINAL SYSTEM RESPONSE (%)	FINAL BIAS (% OF SPAN)	*Acceptable BIAS ± 5%*	MONITOR DRIFT (% OF SPAN)	*Acceptable Drift ≤ 3%*
1	120.00	-2.4	*P*	121.00	-2.0	*P*	0.4	*P*	-1.00	0.0	*P*	0.00	0.4	*P*	0.4	*P*
2	120.00	-2.4	*P*	121.00	-2.0	*P*	0.4	*P*	-1.00	0.0	*P*	0.00	0.4	*P*	0.4	*P*
3	120.00	-2.4	*P*	121.00	-2.0	*P*	0.4	*P*	-1.00	0.0	*P*	0.00	0.4	*P*	0.4	*P*
4	121.00	-2.0	*P*	121.00	-2.0	*P*	0.0	*P*	0.00	0.4	*P*	0.00	0.4	*P*	0.0	*P*
5	121.00	-2.0	*P*	121.00	-2.0	*P*	0.0	*P*	0.00	0.4	*P*	0.00	0.4	*P*	0.0	*P*
6	121.00	-2.0	*P*	121.00	-2.0	*P*	0.0	*P*	0.00	0.4	*P*	0.00	0.4	*P*	0.0	*P*
7	121.00	-2.0	*P*	120.00	-2.4	*P*	0.4	*P*	0.00	0.4	*P*	-1.00	0.0	*P*	0.4	*P*
8	121.00	-2.0	*P*	120.00	-2.4	*P*	0.4	*P*	0.00	0.4	*P*	-1.00	0.0	*P*	0.4	*P*
9	121.00	-2.0	*P*	120.00	-2.4	*P*	0.4	*P*	0.00	0.4	*P*	-1.00	0.0	*P*	0.4	*P*

SUMMARY OF RUNS

RUN	Raw NOx %-DRY	BIAS CORR. NOx %-DRY	Run Avgs %-DRY		O2 %-DRY	NOx lb/MMBtu	Run Avgs lb/MMBtu		Mw 46.01 lb/lb-mol	Ffactor 9600 dscf/MMBtu
1	140.0	142.8			6.77	0.242				
2	141.0	143.8	143.50	R1	7.58	0.259	0.251	R1		
3	141.0	143.8			7.27	0.253				
4	142.0	144.3			7.12	0.251				
5	148.0	150.4	145.02	R2	5.81	0.239	0.242	R2		
6	138.0	140.3			6.62	0.235				
7	134.0	136.7			5.43	0.212				
8	132.0	134.7	135.71	R3	5.43	0.209	0.206	R3		
9	133.0	135.7			4.52	0.198				

*NOx lb/MMBtu differs slightly due to calculations from truncated data